Effective Chirp Modulation Communication System Design for LEO Satellite IoT

Wenqing Zhao, Haozhe Zhang, Jun Zou, Guangzu Liu, and Feng Shu, *Member, IEEE*

***Abstract*—The high-dynamic Doppler frequency offset presents great challenges to communications in the low-Earth orbit (LEO) satellite Internet of Things (IoT) scenarios, including the high-complexity receiver synchronization algorithm, frequency tracking capability, and high-receiver sensitivity. In this article, we propose an effective communication system architecture based on chirp modulation for LEO satellite IoT with reference to LoRa system. Our approach introduces an improved chirp modulation based on low-rate optimization (LRO) with specified frequency interval and a CRC-assisted decoding scheme. Simulation results show that the proposed optimal LRO factor can improve the spectral efficiency. Furthermore, we propose a low-complexity synchronization algorithm and LRO-based frequency tracking strategy. The simulation results confirm the feasibility of the proposed communication system for LEO satellite IoT, which demonstrates robust performance under high-dynamic Doppler frequency offsets.**

***Index Terms*—Chirp modulation, frequency tracking, low-Earth orbit (LEO) satellite communication, synchronization.**

I. Introduction

IN THE recent years, the connectivity requirements of the terrestrial Internet of Things (IoT) end-devices in remote areas (e.g., deserts, oceans, and forests) have driven the advancement of the low-Earth orbit (LEO) satellite IoT [1]. Compared to the geostationary orbit (GEO) satellite, the LEO satellite communications have the advantages of low-path loss, low-propagation delay, and seamless global coverage, enabling the IoT applications in the aforementioned remote areas, such as smart grids and environmental monitoring [2].

However, the rapid motion of the LEO satellite across the Earth's orbit at altitudes between 250 to 2000 km poses several great challenges for LEO satellite IoT communications. The prime challenge is the high-dynamic Doppler frequency offset, which causes a high-dynamic carrier frequency offset (CFO) combined with the low-precision crystal oscillator to the received signals [3]. At an altitude of 560 km with a carrier frequency of 436.7 MHz, the maximum Doppler offset and dynamic Doppler drift rate can reach 11 kHz and 150 Hz/s, respectively [4]. This challenge places high demands on the Doppler tolerance of the transmitted signal, and the receiver's synchronization algorithm to accurately estimate and track the dynamic Doppler frequency offset. Other challenges include the path loss, multisignal blocking and spaceborne electromagnetic interference which require higher receiver sensitivity and enhanced immunity to interference [5], [6].

The existing LEO satellite communication systems can be categorized into narrowband communication systems designed for low-rate services and broadband communication systems intended for high-rate Internet services [7]. The former, exemplified by Iridium and Orbcomm, adopts low-complexity single-carrier wireless communication technologies. However, single-carrier systems are sensitive to frequency offset and phase noise, and are also susceptible to interference and multipath channel fading. The latter, represented by Starlink, employs the multicarrier system based on orthogonal frequency division multiplexing (OFDM), which utilizes multiple orthogonal subcarriers for communication, significantly enhancing both data rate and system robustness against frequency selective fading caused by multipath channels [8]. Nonetheless, challenges, such as high-signal peak-to-average power ratio (PAPR), and intercarrier interference resulting from Doppler frequency offset persist in multicarrier satellite communication system.

To address the above challenges, several satellite systems employ the spread spectrum technology, including direct sequence code division multiple access (DS-CDMA) and chirp spread spectrum (CSS). DS-CDMA, widely used in satellite communications (e.g., GPS and Globalstar), offers high-network capacity and strong anti-interference capabilities [9]. The PMF-FFT algorithm is typically employed for the DSSS signal's acquisition, but the long spread spectrum sequence and high-dynamic Doppler frequency offset increase the difficulty and complexity of DSSS signal's demodulation [10], [11]. The CSS modulation is originally applied in LoRa system for its wide communication coverage, robustness to the frequency offset and interference. Compared with DSSS signal, the CSS signal is acquired by detecting successive chirp synchronization signals, which reduces computation complexity by eliminating the sliding correlation. Due to these advantages, the CSS-based LoRa is emerging as a promising solution for enabling LEO satellite IoT, providing seamless global coverage [12]. Besides, LEO satellite communication can benefit from the combination with reconfigurable intelligent

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surface (RIS) [13], which can intelligently enhance the wireless propagation environment [14], [15], and can also be configured as a transmitter to realize information transmission with low-hardware complexity [16].

Researchers are paying close attention to the adaptability of CSS-based LoRa in LEO satellite IoT scenarios and related improvement initiatives. The adaptability of CSS modulation and the LoRaWAN protocol, including aspects like the network architecture and access mechanism, in LEO satellite IoT scenarios have been analyzed in [17] combined with the characteristics of the LEO satellite. Furthermore, the aspects related to link budget and packet collision in LoRa direct to satellite (DtS) link are considered in [18] and [19], which demonstrate the potential feasibility of LoRa DtS. Asad Ullah et al. [4] analyzed the impact of high-dynamic Doppler frequency offset on LoRa DtS performance under various communication parameter settings, e.g., bandwidth, carrier frequency, and MAC payload by modeling the Doppler offset experienced in the LEO satellite uplink.

To address the challenges posed by the multiple access, network capacity and dynamic Doppler frequency offset, the CSS modulation has recently been improved in the literature. In [20], symmetry CSS (SCSS) modulation is proposed to reduce the maximum cross-correlation and enable multiple access communication in LEO satellite IoT. In [21], interleaved chirp spreading LoRa (ICS-LoRa) is proposed to enhance the LoRa network capacity and improve the data rate by expanding the signal dimensions. Furthermore, folded chirp-rate shift keying (FCRSK) modulation is proposed in [22] with stronger robustness to the dynamic Doppler frequency offset than CSS modulation, but at the cost of lower spectral efficiency and noise immunity. The number of information bits modulated at a LoRa symbol is proposed to be adapted in [23], which increases the frequency interval between the adjacent effective frequency indexes to enhance the robustness of LoRa to the dynamic Doppler frequency offset.

Apart from the improvement of CSS modulation, accurate estimation of CFO and symbol timing are critical for reliable communication in LEO satellite IoT. The CFO is decomposed into static CFO (SCFO) originating from Doppler offset and low-precision crystal oscillators, and dynamic CFO (DCFO) caused by dynamic Doppler drift rate. Several effective synchronization algorithms have recently been proposed for terrestrial LoRa system which focus on the estimation of SCFO and symbol timing error (STO). In [24], a low-complexity synchronization algorithm is proposed for SCFO and STO estimation, but the proposed estimator greatly degrades the PER performance of LoRa. To address this issue, an iterative estimator for the fractional STO is proposed in [25], which significantly improves the PER performance under SCFO below 20 ppm. The estimator of the fractional STO in [25] relies on the accurate estimation of fractional SCFO, which is performed by calculating the phase difference between the spectral peaks demodulated from the adjacent synchronization signals. However, the fractional SCFO estimator employed in [24] and [25] will be ineffective in the presence of DCFO. The PER improvement observed in [25] is compromised when SCFO exceeds 20 ppm, typical in LEO satellite communication.

Additionally, the DCFO tracking remains pivotal in LEO satellite communication. The improvement initiatives in [22] and [23], including the low-data-rate optimization (LDRO) in [4], have not addressed the DCFO tracking issue. Traditional DCFO tracking methods include the extended Kalman filter, cross-product automatic frequency control loop and the phase-locked loop with additional computation complexity [26]. A sampling frequency offset (SFO) tracking method for CSS receivers is mentioned in [27], which extract the SFO error by using FFT output and then control digital PLL to track SFO, but bringing high complexity.

Generally, the CSS-based LoRa communication technology shows potential for LEO satellite IoT applications. However, its direct implementation poses several great challenges. The standalone CSS modulation is susceptible to residual CFO and STO, and the DCFO tracking issue is not considered in current LoRa system, while the traditional DCFO tracking methods are complex to be applied to LEO satellite scenarios. Additionally, the existing synchronization strategies for LoRa receiver are designed for terrestrial IoT, without considering DCFO. Motivated by these challenges, this article investigates the chirp modulation communication system design for LEO satellite IoT, which addresses the above challenges with low complexity.

The contributions and innovations of this article are as follow.

1) In this work, we propose an effective chirp modulation communication system architecture for LEO satellite IoT and show the design details of the transceivers. An improved chirp modulation scheme based on low-rate optimization (LRO), and a CRC-assisted decoding scheme are proposed to enhance the system's robustness against residual CFO and STO. Furthermore, the optimal value of the LRO factor is proposed to improve the spectral efficiency.
2) We model the received signal contaminated with CFO and STO under AWGN and analyze the impact of both the integer and fractional components of SCFO and STO on the chirp signal's spectrum and detection.
3) A low-complexity synchronization algorithm is proposed to estimate both the SCFO and STO. Furthermore, we propose an LRO-based DCFO tracking strategy with low-computation cost and analyze the tracking capability of the proposed strategy in detail.
4) The proposed chirp modulation system can be easily implemented by FPGA or DSP according to the detailed signal processing procedure in this article. It can be treated as a tutorial of chirp-based communication system design. Currently, four satellites of Tianqi Constellation in orbit have utilized the chirp-based communication system proposed in this article, and the system is functioning well in space.

The remainder of this article is organized as follows. The transmitter design, including the frame structure design, coding scheme, framing and LRO-based chirp modulation scheme, is described in Section II. The receiver design, including the

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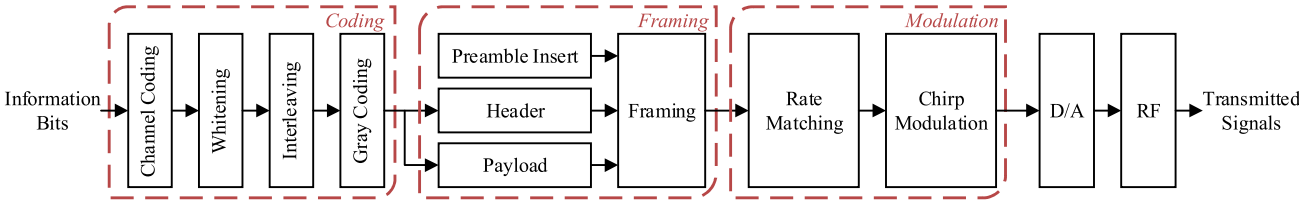


Fig. 1. Architecture of the transmitter based on chirp modulation.

received signal model, synchronization algorithm, LRO-based DCFO tracking strategy, and CRC-assisted decoding scheme, as well as the receiver complexity analysis, is provided in Section III. The simulation results of the proposed system are presented in Section IV and the conclusion of this article is drawn in Section V. The major variables in this article are listed in Table I for ease of reference.

II. Design of the Transmitter

In this section, we present the transmitter design based on chirp modulation, specifically designed for LEO satellite IoT with high-dynamic Doppler frequency offset. This section focuses on the design details of the frame structure, coding scheme, framing and LRO-based chirp modulation scheme.

A. Frame Structure

The architecture of the proposed transmitter is shown in Fig. 1 which consists of coding, framing and modulation. In the time domain of the proposed system, it can be divided into preamble, header and payload, as depicted in Fig. 2.

The preamble is applied in the symbol synchronization and comprises three parts. The first part consists of m_0 upchirps, i.e., chirps with a linear-increasing frequency, modulated with a constant value of $a_0 = 0$. The second part consists of m_1 frame synchronization signals (FSSs) modulated with the transmitter's network ID ($a_1 \sim a_{m_1}$), used for the verification and synchronization of the received frame. The number of FSSs is configurable according to the selected system parameters to reduce the false alarm probability. The last part includes m_2 downchirps, i.e., chirps with a linear-decreasing frequency, also modulated with $a_0 = 0$. The combination of the successive upchirps and downchirps facilitates the fast acquisition of the chirp signal and enables the estimation of SCFO and STO in the receiver.

Assuming that the chirp symbol period is T_s, the additional downchirp with duration $T_s/4$ is attached at the beginning of the header which corresponds to the maximum correctable STO. The additional downchirp is inserted for the ease of hardware implementation of timing adjustment. Furthermore, the preamble length is changeable depending on the application scenarios. The short preamble helps to enhance the system efficiency, whereas the long preamble can be employed to wake up dormant devices and reduce the false alarm probability.

As shown in Fig. 3, the header is employed to instruct the crucial parameters of the transmitted data frame, including the payload length (8-bits), data domain CRC option (1-bit), and coding rate (3-bits). Three reserved bits in the header facilitate protocol upgrades, enabling instruction for the LRO option and channel coding scheme. Additionally, the 5-bits header CRC is employed to check the header data, which helps to reduce the false alarm probability. The last segment of the frame is the data and its CRC.

TABLE I
Variables List

Notation	Definition
m_0, m_1, m_2	Numbers of upchirps, FSSs and downchirps in the preamble
a_0, $a_1 \sim a_{m_1}$	Synchronization word and transmitter network ID
T_s	Chirp symbol period
$\mathbf{b}$, $\mathbf{p}$, $\mathbf{b}'$	Information bits, data domain CRC and CRC-coded information bits
R, C_B, C_F	Coding rate, number of parity bits, and coding factor
$\mathbf{S}$, $\mathbf{K}$	Transmitted symbol sequence and valid frequency index sequence
SF, l	Spreading factor and LRO factor
K_i, $\hat{K}_i$	Modulated valid frequency index and demodulated frequency index
μ, B, T_s, N	Chirp signal frequency modulation slope, bandwidth and time-bandwidth product
$s(t)$, $r(t)$	Transmitted signal and received signal
$y(n)$, $Y(k)$	Dechirped signal and DFT output
τ, $\Delta f_c(t)$	STO and CFO
K_τ, K_{τ_α}, K_{τ_β}	Normalized STO, integer and fractional part of normalized STO
K_f, K_{f_α}, K_{f_β}	Normalized SCFO, integer part and fractional part of normalized SCFO
$r_i(n)$, n_i^s	Detected chirp signal and timing factor
f_i^e, ρ	Frequency correction factor and fixed adjustment factor
$\hat{r}_k$, $\mathbf{k}$	Parity bit and index set of erroneous codewords
$\hat{\mathbf{b}}$, $\hat{\mathbf{p}}$, $\Delta P(X)$	Recovered information bits, data domain CRC and CRC checking polynomial
$F_v(X)$, v_{z,s_z}	Error pattern polynomial and its index

B. Coding Scheme

The generalized coding scheme is proposed to correct the transmission errors induced by the channel, encompassing the channel coding, whitening, interleaving, and gray coding.

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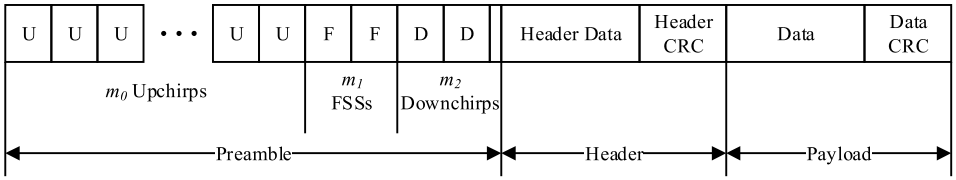


Fig. 2. Frame structure of the proposed system.

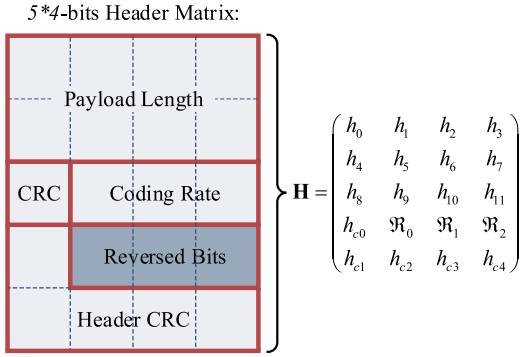


Fig. 3. Header structure of the proposed system where a square denotes a header bit and $\mathfrak{R}_i$ denotes the reversed bit for $i \in [0, 2]$.

The header data and the information bits are appended with the header CRC and data domain CRC, respectively. The header CRC is generated using the header CRC generator matrix $\mathbf{G_{hc}}$ which can be expressed as

$$\mathbf{h_c} = \mathbf{h} \cdot \mathbf{G_{hc}} \bmod 2 = (h_{c0}, h_{c1}, h_{c2}, h_{c3}, h_{c4}) \tag{1}$$

where $\mathbf{h} = (h_0, h_1, \ldots, h_{11})$ is the 12-bits header data. The CRC generator polynomial

$$g_{\text{crc16}}(X) = X^{16} + X^{12} + X^5 + 1 \tag{2}$$

is employed to generate the 16-bits data domain CRC, denoted as $\mathbf{p} = (p_0, p_1, \ldots, p_{15})$. The information bits with length of L, denoted as $\mathbf{b} = (b_0, b_1, \ldots, b_{L-1})$, are appended with the data domain CRC, forming the CRC-coded bit sequence

$$\mathbf{b}' = (\mathbf{b}, \mathbf{p}) = (b_0, b_1, \ldots, b_{L-1}, p_0, p_1, \ldots, p_{15}). \tag{3}$$

Various channel coding schemes can be employed in this scheme, such as Hamming codes, LDPC, polar codes, turbo codes, and so on. In this work, the low-complexity Hamming coding scheme is chosen for channel coding as an example. We propose harnessing the error detection capability of the $(5, 4)$ Hamming codes and potential CRC error patterns to detect the indexes of error bits, as detailed in Section III. The Hamming codeword is generated using a Hamming generator matrix $\mathbf{G}_R$, which correlates with the coding rate $R = 4/(4 + C_B)$, where C_B is the number of parity bits in a codeword. The kth codeword after Hamming coding can be expressed as

$$\mathbf{c}_k = \mathbf{u}_k \cdot \mathbf{G}_R \bmod 2 = \left(c_{k,0}, c_{k,1}, \ldots, c_{k,3+C_B}\right) \tag{4}$$

where $\mathbf{u}_k = (b'_{4k}, b'_{4k+1}, b'_{4k+2}, b'_{4k+3})$ is the information segment of the kth coding block.

A whitening sequence is added to the Hamming codeword in GF(2) to randomize the transmitted bit sequence which can be written as

$$\mathbf{d}_k = (\mathbf{c}_k + \mathbf{w}_k) \bmod 2 = \left(d_{k,0}, d_{k,1}, \ldots, d_{k,3+C_B}\right) \tag{5}$$

where $\mathbf{w}_k = (w_{k,0}, w_{k,1}, \ldots, w_{k,3+C_B})$ is the whitening sequence and $\mathbf{d}_k$ is the kth whitened codeword.

Diagonal interleaving is introduced to enhance the proposed system's robustness against consecutive bit errors caused by the impulse interference in the channel. The whitened bit sequence, denoted as $\mathbf{d}$, is sequentially written into the interleaving block row-wise which can be written as

$$\mathbf{X} = \begin{pmatrix} \mathbf{d}_k \\ \mathbf{d}_{k+1} \\ \vdots \\ \mathbf{d}_{k+C_F-1} \end{pmatrix} = \begin{pmatrix} d_{k,0} & d_{k,1} & \cdots & d_{k,3+C_B} \\ d_{k+1,0} & d_{k+1,1} & \cdots & d_{k+1,3+C_B} \\ \vdots & \vdots & \ddots & \vdots \\ d_{k+C_F-1,0} & d_{k+C_F-1,1} & \cdots & d_{k+C_F-1,3+C_B} \end{pmatrix} \tag{6}$$

where $\mathbf{X}$ is the interleaving block with size of $C_F \times (3 + C_B)$ and C_F is defined as the *coding factor*. The whitened bits in (6) are diagonally interleaved based on the following rule:

$$\mathbf{X}'(\xi, \eta) = \mathbf{X}(C_F - 1 - ((\eta - \xi) \bmod C_F), 3 + C_B - \xi) \quad \text{for } \xi \in [0, 3 + C_B], \eta \in [0, C_F - 1] \tag{7}$$

where $\mathbf{X}'$ is the interleaved block and ξ, η are the row and column indexes of $\mathbf{X}'$, respectively. The interleaved bits in $\mathbf{X}'$ are described as follows:

$$\mathbf{X}' = \begin{pmatrix} \mathbf{e}_0 \\ \mathbf{e}_1 \\ \vdots \\ \mathbf{e}_{3+C_B} \end{pmatrix} = \begin{pmatrix} d_{k+C_F-1,3+C_B} & d_{k+C_F-2,3+C_B} & \cdots & d_{k,3+C_B} \\ d_{k,2+C_B} & d_{k+C_F-1,2+C_B} & \cdots & d_{k+1,2+C_B} \\ \vdots & \vdots & \ddots & \vdots \\ d_{k+2+C_B,0} & d_{k+1+C_B,0} & \cdots & d_{k+3+C_B,0} \end{pmatrix} \tag{8}$$

where $\mathbf{e}_i$ is the ith interleaved bit sequence with length of C_F, which can be written as

$$\mathbf{e}_i = \left[d_{k+J_i^0,3+C_B-i}, d_{k+J_i^1,3+C_B-i}, \ldots, d_{k+J_i^{C_F-1},3+C_B-i}\right]. \tag{9}$$

Here, $J_i^j = (C_F - 1 - j + i) \bmod C_F$ with bit index $j \in [0, C_F)$. The diagonal interleaving process is shown in Fig. 4, which can increase the data dispersion compared to block interleaving and spread errors to the different chirp symbols.

Gray coding is then applied to the interleaved bit sequence $\mathbf{e}_i$, which is padded with $16 - C_F$ zeros to form the 16-bits sequence $\boldsymbol{\mu}_i = (\mu_{i,0}, \mu_{i,1}, \ldots, \mu_{i,15})$ and subsequently coded according to the gray coding rule

$$\boldsymbol{\mu}'_i(\upsilon) = \begin{cases} \mu'_{i,\upsilon} = \left(\mu'_{i,\upsilon-1} + \mu_{i,\upsilon}\right) \bmod 2, & \text{for } \upsilon \neq 0 \\ \mu'_{i,0} = \mu_{i,0}, & \text{for } \upsilon = 0 \end{cases} \tag{10}$$

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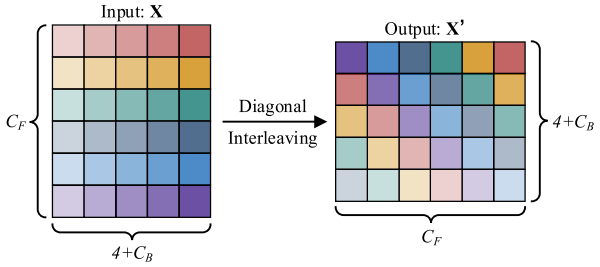


Fig. 4. Visualization of the diagonal interleaving with an $C_F \times (4 + C_B)$ interleaving block where $C_F = 6$ and $C_B = 1$.

where $\upsilon \in [0, 15]$ is the gray coding index. The Gray-coded bit sequence $\boldsymbol{\mu}'_i$ is finally converted into a decimal symbol $C_i = \sum_{\upsilon=0}^{15} 2^{15-\upsilon} \mu'_{i,\upsilon}$.

C. Framing

As depicted in Fig. 2, the first segment of the proposed data frame is the preamble. The symbol sequence modulated at the preamble is represented as $\mathbf{A} = (A_0, A_1, \ldots, A_{N_1-1})$ where $N_1 = m_0 + m_1 + m_2$. Specifically, the ith symbol, denoted as A_i, can be expressed as

$$A_i = \begin{cases} a_0, & \text{for } i \in [0, m_0 - 1] \bigcup [m_0 + m_1, N_1 - 1] \\ a_{i-m_0+1}, & \text{for } i \in [m_0, m_0 + m_1 - 1]. \end{cases} \tag{11}$$

The second segment of the data frame is the header, with a row number of 5 as depicted in Fig. 3. Partial coded and whitened information bits with size of $(C_F - 5) \times (C_B + 4)$ are added to the header in cases where $C_F > 5$ to facilitate the diagonal interleaving of the header where the row number of the interleaving block is equal to C_F as depicted in Fig. 4. The symbol sequence generated by the header is denoted as $\mathbf{B} = (B_0, B_1, \ldots, B_{N_2-1})$ where $N_2 = 4 + C_B$.

The last segment of the data frame is the data and data domain CRC. Except the complementary information bits added to the header, the remaining whitened information bits are diagonally interleaved and Gray-coded to generate the symbol sequence, denoted as $\mathbf{C} = (C_0, C_1, \ldots, C_{N_3-1})$ where

$$N_3 = \left\lceil \frac{L + 16 - 4(C_F - 5)}{4 \times C_F} \right\rceil \times (4 + C_B) \tag{12}$$

is the length of $\mathbf{C}$. The transmitted symbol sequence, denoted as $\mathbf{S}$, is formed by sequentially combining the aforementioned three symbol sequences with length of $N_s = N_1 + N_2 + N_3$, which can be expressed as

$$\mathbf{S} = [\mathbf{A}, \mathbf{B}, \mathbf{C}]. \tag{13}$$

D. LRO-Based Chirp Modulation

The proposed LRO-based chirp modulation is composed of the rate matching and chirp modulation. LRO is introduced to enhance the system's robustness against residual CFO and STO by increasing the protective interval between adjacent valid frequency indexes, performed by the rate matching.

During the rate matching, the transmitted symbol sequence $\mathbf{S}$ is mapped to the valid frequency index sequence denoted as $\mathbf{K} = [K_0, K_1, \ldots, K_{N_s-1}]$ which obeys the following mapping rule:

$$K_i = \begin{cases} S_i, & \text{for } i \in [0, N_1 - 1] \\ l \times S_i + 1, & \text{for } i \in [N_1, N_s - 1] \end{cases} \tag{14}$$

where l denotes the *LRO factor*. Notably, LRO is not applied in the preamble. Here, $S_i \in [0, 2^{C_F-1}]$ is the ith transmitted symbol consisting of C_F effective bits, while $K_i \in [0, 2^{\mathrm{SF}} - 1]$ is the ith valid frequency index composed of SF bits, where SF is defined as the *spreading factor*.

If LRO is not applied in the modulation, the mapping rule (14) is adjusted to

$$K_i = S_i, \text{ for } i \in [0, N_s - 1] \tag{15}$$

which requires $C_F = \mathrm{SF}$. This implies that all the frequency indexes are valid and each index carries C_F effective bits.

The chirp modulation without LRO is vulnerable to residual CFO and STO. The chirp signal demodulation involves performing a 2^{SF}-point DFT on the processed chirp signal and searching for the spectral peak to determine the modulated frequency index, which is specifically described in Section III. In the demodulation, the residual CFO and STO introduce an offset to the spectral peak and thus the obtained frequency index deviates from the modulated frequency index. As depicted in the top of Fig. 5, the solid orange arrow depicts the frequency index offset due to the residual CFO and STO, while the dotted orange arrow depicts the index demodulation under the picket-fence effect of the 2^{SF}-point DFT. The frequency interval of the modulation without LRO is $1 \cdot \Delta F$ where $\Delta F = f_s/2^{\mathrm{SF}}$ is the minimal frequency interval of the 2^{SF}-point DFT, with f_s representing the sampling rate. Without considering AWGN, index demodulation error occurs when the spectral peak offset exceeds $(-\Delta F/2, \Delta F/2]$.

If LRO is applied in the modulation as described in (14), the frequency interval between adjacent valid frequency indexes is expanded to $l \cdot \Delta F$, which provides the capability to correct the index demodulation errors at the receiver. As illustrated in the lower half of Fig. 5, gray indicates invalid frequency indexes, the translucent region represents the correction range of each valid frequency index, and the dotted green arrow

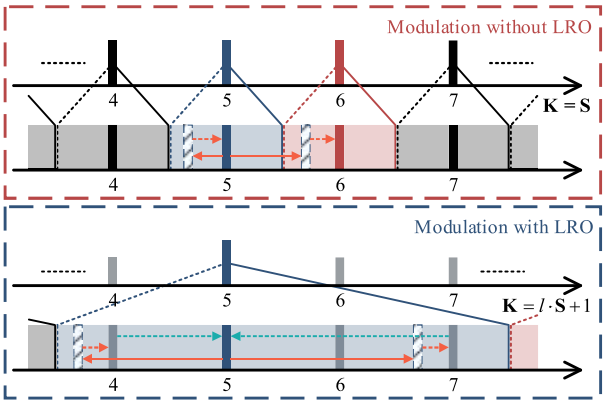


Fig. 5. Comparisons of the modulation with LRO and without LRO, and corresponding index demodulation in the presence of residual CFO and STO under $l = 4$.

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signifies the correction of index demodulation errors. When the frequency index K_i is wrongly demodulated as $\widehat{K}_i \in \{K_i \pm 1, K_i + 2\}$, the demodulation error can be corrected to $\widehat{K}_i' = K_i$ with $l = 4$. Facilitated by the expanded frequency interval, the invalid frequency index can be utilized to track the DCFO which is proposed in Section III. Compared to the modulation without LRO, the number of the valid frequency indexes is reduced to $\lfloor 2^{\mathrm{SF}}/l \rfloor$ and thus the number of effective bits carried by each valid frequency index is reduced to $C_F = \mathrm{SF} - \log_2 l$, which leads to a reduction in spectral efficiency.

In the LRO operation with $l = 4$, the middle invalid index between adjacent valid frequency indexes cannot effectively correct index demodulation errors due to its equidistance from adjacent valid indexes. To improve spectral efficiency, the optimal LRO factor is advisable to be selected as $l = 3$. However, this LRO factor results in $C_F = \mathrm{SF} - \log_2 3$, where the valid frequency index should carry fractional effective bits. Consequently, three consecutive valid frequency indexes are combined to carry $\lfloor \log_2(2^{3\mathrm{SF}}/27) \rfloor$ effective bits. During the coding stage, C_F is adjusted to SF in order to facilitate the diagonal interleaving. Furthermore, the M-ary numbering system is utilized to map the coded bit sequence to the symbol sequence, where $M = \lfloor 2^{\mathrm{SF}}/l \rfloor$ serves as the base number. Unfortunately, gray coding cannot be employed in the M-ary numbering system because the frequency index mapped from the Gray-coded M-ary number may exceed the maximum frequency index $K_{\max} = 2^{\mathrm{SF}} - 1$.

The proposed system employs chirp signal as the modulated signal, characterized by linear frequency variation over the symbol duration T_s. The frequency modulation slope of the chirp signal is $\mu = B/T_s$, where B is the chirp signal's bandwidth. Additionally, the time-bandwidth product of the chirp signal is $N = \mathrm{BT}_s = 2^{\mathrm{SF}}$.

The frequency index generated by LRO is modulated at the initial frequency of the chirp, known as frequency shift chirp modulation (FSCM) [28]. The continuous-time expression of an upchirp modulated with the frequency index K_i is

$$x_{\mathrm{up}}(t, K_i) = e^{j2\pi\left(\frac{\mu}{2}t^2 + \left(\frac{K_i}{N} - \frac{1}{2} - u(t - t_f)\right)Bt\right)}, \ \text{for } 0 \le t \le T_s \quad (16)$$

where $u(t)$ is the step function, and $t_f = (N - K_i)/B$ is the frequency folding time at which the instantaneous frequency is decreased to $-B/2$. The upchirp is sampled at Nyquist rate $f_s = B$ to obtain its discrete-time expression, expressed as

$$x_{\mathrm{up}}(n, K_i) = e^{j2\pi\left(\frac{n^2}{2N} + \left(\frac{K_i}{N} - \frac{1}{2}\right)n\right)}, \ \text{for } 0 \le n \le N - 1. \quad (17)$$

The downchirp signal, denoted as $x_{\mathrm{down}}(n, K_i)$, is the complex conjugation of the upchirp, as described in (17). The ith chirp signal of the framed signal can be expressed as

$$s_i(n) = \begin{cases} x_{\mathrm{down}}(n, K_i), & \text{for } m_0 + m_1 \le i < N_1 \\ x_{\mathrm{up}}(n, K_i), & \text{others.} \end{cases} \quad (18)$$

The framed signal is sequentially composed of the chirps modulated with corresponding frequency indexes which can be expressed as

$$s(n) = \sum_{i=0}^{N_s - 1} s_i\left(n - iN^*\right), \ \text{for } n \in \left[0, N_s N^* - 1\right] \quad (19)$$

where N^* is the number of sampling points per symbol period, expressed as

$$N^* = \begin{cases} N/4, & \text{for } i = N_1 - 1 \\ N, & \text{others.} \end{cases} \quad (20)$$

The framed signal is then converted to an analog signal $s(t)$ by DAC where $s(t)$ is a continuous-phase signal because

$$\varphi_i(0) = \varphi_i(T_s) = 0, \ \varphi_i\left(t_f^-\right) = \varphi_i\left(t_f^+\right) = \frac{\pi K_i(N - K_i)}{N}. \quad (21)$$

Here, $\varphi_i(t)$ is the instantaneous phase of $x_{\mathrm{up}}(t, K_i)$, and this property is consistent in $x_{\mathrm{down}}(t, K_i)$. The analog signal $s(t)$ is finally transmitted through the RF front end.

III. Design of the Receiver

In this section, we present the demodulation method for the chirp signal and analyze the received signal which is modeled contaminated by CFO and STO under an AWGN channel. The design details of the receiver are then provided, including the low-complexity synchronization algorithm, LRO-based DCFO tracking strategy, and CRC assisted decoding scheme.

A. Demodulation for the Chirp Signal

In the demodulation process, the chirp signal described in (17) is first multiplied by a local sequence, denoted as $c(n)$, where this multiplication operation is called as *dechirp*. The dechirped signal can be written as

$$y(n) = x_{\mathrm{up}}(n, K) \cdot c(n), \ \text{for } n \in [0, N - 1] \quad (22)$$

where K denotes the frequency index modulated on the upchirp signal. The local sequence $c(n)$ depends on the chirp signal to be dechirped and can be expressed as

$$c(n) = \begin{cases} e^{-j2\pi\left(\frac{n^2}{2N} - \frac{n}{2}\right)}, & \text{for upchirp} \\ e^{j2\pi\left(\frac{n^2}{2N} - \frac{n}{2}\right)}, & \text{for downchirp.} \end{cases} \quad (23)$$

The chirp signal demodulation is achieved by performing an N-point DFT on the dechirped signal. The DFT output of the dechirped signal in (22) can be written as

$$Y(k) = \sum_{n=0}^{N-1} y(n) e^{-j\frac{2\pi}{N}kn} = N\delta(k - K), \ \text{for } k \in [0, N - 1] \quad (24)$$

where $\delta(n)$ denotes the Kronecker function. The frequency index modulated at the upchirp signal can be obtained by searching for the frequency index with the greatest magnitude [29], which can be described as

$$\widehat{K} = \arg\max_k |Y(k)|. \quad (25)$$

B. Received Signal Modeling

The received signal contaminated by CFO and STO under AWGN channel can be written as

$$r(t) = s(t + \tau) e^{j2\pi\Delta f_c(t)t} + w(t) \quad (26)$$

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where τ and $\Delta f_c(t)$ denote STO and CFO, respectively. Here, $w(t) \sim \mathrm{CN}(0, \sigma^2)$ is the complex white Gaussian noise with zero-mean and variance σ^2. Based on this model, the received upchirp with $K = 0$ can be expressed as

$$r_u(t) = x_{\mathrm{up}}(t, 0) \cdot \underbrace{e^{j2\pi(\Delta f_c(t)+\mu\tau)t}}_{\mathrm{I}} \cdot \underbrace{e^{-j2\pi B(t+\tau)u(t-T_s+\tau)}}_{\mathrm{II}} \cdot \underbrace{e^{j\varphi_u}}_{\mathrm{III}} + w(t) \quad \text{for } t \in [0, T_s) \tag{27}$$

where term-I denotes the frequency offset introduced by CFO and STO, and term-II represents the frequency folding at the boundary between two consecutive upchirps. In term III, $\varphi_u = \pi(\mu\tau - B)\tau$ is the constant additional phase introduced by STO which does not affect the demodulation described in (25) and is therefore omitted in the subsequent analysis.

The received upchirp signal as described in (27) is sampled at Nyquist rate $f_s = B$, where STO and the static part of CFO (SCFO) are decomposed into the integer and fractional parts

$$\tau = \frac{K_\tau}{B} = \frac{K_{\tau_\alpha} + K_{\tau_\beta}}{B} \tag{28}$$

$$\Delta f_c = K_f \cdot \frac{B}{N} = \left(K_{f_\alpha} + K_{f_\beta}\right) \cdot \frac{B}{N}. \tag{29}$$

Here, $K_{\tau_\alpha}, K_{f_\alpha}$ are the integer parts, whereas $K_{\tau_\beta}, K_{f_\beta} \in [-0.5, 0.5)$ are the fractional parts. DCFO is addressed by the LRO-based tracking strategy, which is thus omitted in this analysis. Furthermore, the discrete-time expression of the received upchirp signal can be written as

$$r_u(n) = x_{\mathrm{up}}(n, 0) e^{j2\pi\left(\frac{K_f+K_\tau}{N} n - K_{\tau_\beta} u(n-N_b)\right)} + w(n) \tag{30}$$

where $N_b = N - \lfloor K_\tau \rfloor$ denotes the boundary between two upchirps. This formulation demonstrates that both SCFO and STO contribute to the frequency offset, while the fractional STO introduces a phase discontinuity of $-2\pi K_{\tau_\beta}$ at the boundary between two consecutive upchirps [25].

The received upchirp, as described in (30), is demodulated using the method outlined in (22)–(25). In the presence of residual SCFO and STO, the DFT result of the dechirped signal excluding the consideration of AWGN, as described in (24), can be rewritten as

$$Y_u(k) = \left(1 - e^{-j2\pi K_{\tau_\beta}}\right) \Pi\left(\frac{N_b}{N}\left(k - K_f - K_\tau\right), N_b\right) + e^{-j2\pi K_{\tau_\beta}} \Pi\left(k - K_f - K_\tau, N\right), \quad \text{for } k \in [0, N-1] \tag{31}$$

where $\Pi(x, y) \triangleq (1 - e^{-j2\pi x})/(1 - e^{-j2\pi[x/y]})$ is the discrete *sinc function*. The spectra of the dechirped signals with varying SCFOs and STOs are presented in Fig. 6. The fractional SCFO and STO result in attenuation and shift in the spectral peak, accompanied by spectral leakage and the picket-fence effect. The attenuation of the spectral peak is positively correlated with the fractional values of SCFO and STO.

C. Synchronization Algorithm

As illustrated in Fig. 7, the proposed receiver is sequentially composed of synchronization, demodulation and decoding. The proposed synchronization algorithm is described in Algorithm 1, which can be divided into three stages, corresponding to the detection of the upchirps, FSSs and downchirps, respectively.

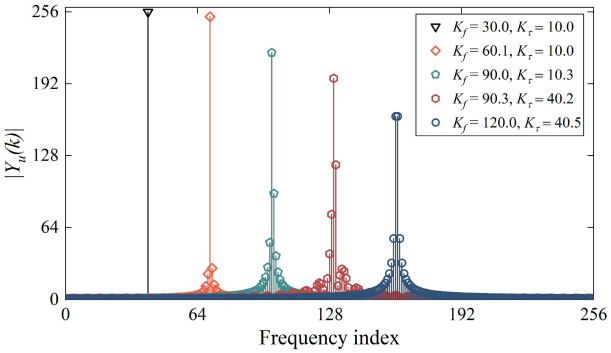


Fig. 6. Spectra of the dechirped signals with varying SCFOs and STOs.

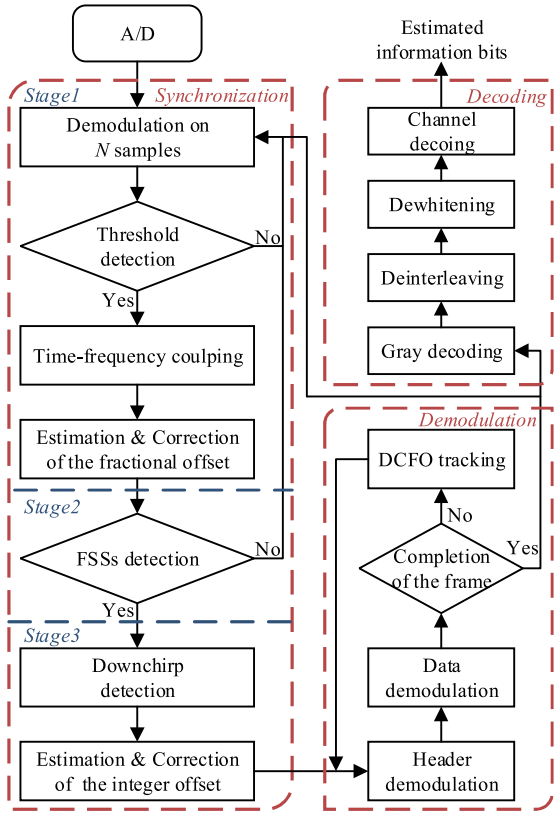


Fig. 7. Architecture and workflow of the proposed receiver.

The chirp signal detection is processed by an N-point sliding detection window, with step length N adjusted by the timing factor, denoted as n_i^s. Consequently, the chirp signal in the ith detection window is

$$r_i(n) = r\left(n + n_i^s + (i-1) \cdot N\right), \quad \text{for } n \in [0, N-1] \tag{32}$$

where the timing factor n_i^s depends on the synchronization stage, specifically described in (37).

1) Stage1 (Upchirp Detection): As depicted in Fig. 7, the purpose of the upchirp detection is to confirm the arrival of a data frame through the threshold detection. The chirp signals potentially detected during this stage are consecutive

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Algorithm 1 Proposed Synchronization

$i = 0$, $n_i^s = 0$
Stage1 Upchirp Detection:
$i = i + 1$, $n_i^s = n_{i-1}^s$
$\widehat{K}_i = \arg\max_k |\mathrm{DFT}[r_i(n) \cdot c(n)]|$
Compute P_s^i, $\overline{P}_n^i$ by (33) and (34)
if $(P_s^i/\overline{P}_n^i \geq P_t)$, $I_u = i$, goto step 8
else repeat step 2∼6
end if
$i = i + 1$, $n_i^s = n_{i-1}^s - (\widehat{K}_{i-1} - K_{i-1})$, compute f_i^e by (43)
$\widehat{K}_i = \arg\max_k |\mathrm{DFT}[r'_i(n) \cdot c(n)]|$
if $(P_s^i/\overline{P}_n^i < P_t)$, goto step 1
else if $(|\gamma(\widehat{K}_i, a_0)| > \Delta_1$ && $|\gamma(\widehat{K}_i, a_1)| \leq \Delta_2)$
  $I_F = i$, $i = i + 1$, $I_d = I_F + m_1$, goto step 15
else repeat step 8∼13
end if

Stage2 FSS Detection:
for $i = I_F + 1$ to $I_d - 1$ **do**
$n_i^s = n_{i-1}^s - (\widehat{K}_{i-1} - K_{i-1})$, $f_i^e = f_{i-1}^e$
$\widehat{K}_i = \arg\max_k |\mathrm{DFT}[r'_i(n) \cdot c(n)]|$
if $(P_s^i/\overline{P}_n^i < P_t || |\gamma(\widehat{K}_i, a_{i-I_F+1})| > \Delta_2)$, goto step 1
  else repeat step 15∼19
end if
end for

Stage3 Downchirp Detection:
for $i = I_d$ to $I_d + m_2 - 2$ **do**
  $n_i^s = n_{i-1}^s$, $f_i^e = f_{i-1}^e$
  $\widehat{K}_i = \arg\max_k |\mathrm{DFT}[r'_i(n) \cdot c(n)]|$
end for
Estimate $\widehat{K}_{f_\alpha}$ and $\widehat{K'}_{\tau_\alpha}$ by (51)
Realign symbol timing and correct frequency offset

m_0 upchirps modulated with the same frequency index $a_0 = 0$. Consequently, the chirp signal in the ith detection window, denoted as $r_i(n)$, is as described in (30). The frequency index of the spectral peak, denoted as $\widehat{K}_i$, can be obtained by demodulating $r_i(n)$ according to the method outlined in (22)–(25).

In this detection, the power of the dechirped signal $y_i(n)$ is

$$P_s^i = \left|Y_i\left(\widehat{K}_i\right)\right|^2 \tag{33}$$

where $Y_i(k)$ is the spectrum of $y_i(n)$. The average noise power is calculated as

$$\overline{P}_n^i = \frac{\sum_{k=0}^{N-1} |Y_i(k)|^2 - P_s^i}{N-1}. \tag{34}$$

A successful upchirp detection is conditioned on

$$P_s^i/\overline{P}_n^i \geq P_t \tag{35}$$

where P_t is the detection threshold. The successful threshold detection confirms the arrival of a data frame and I_u is defined as the arrival flag of the data frame. This threshold detection is consistently applied in subsequent chirp signal detections. If the threshold detection fails for $i > I_u$, the data frame detection is considered false and the receiver restarts the upchirp detection process.

In the I_uth detection, the frequency index of the spectral peak, without considering AWGN, can be expressed as

$$\begin{aligned}\widehat{K}_{I_u} &= \left(\psi\left(K_f + K_\tau\right)\right) \bmod N \\ &= \left(K_{f_\alpha} + K_{\tau_\alpha} + \psi\left(K_{f_\beta} + K_{\tau_\beta}\right)\right) \bmod N\end{aligned} \tag{36}$$

where $\psi(x)$ denotes the rounding operation and $\widehat{K}_{I_u}$ reflects the shift of the spectral peak from the modulated frequency index $K_{I_u} = a_0$, caused by SCFO and STO. *Time-frequency coupling* is defined to represent the scenario where the shifts caused by SCFO and STO are equal in value but opposite in direction, i.e., $K_f + K_\tau = 0$. This scenario can simplify the estimation of SCFO and STO in the presence of DCFO, and facilitate the demodulation of FSSs.

To achieve the time-frequency coupling for the received signal, the receiver skips specified samples during the upchirp and FSS detection by adjusting the timing factor, which can be specifically described as

$$n_{i+1}^s = \begin{cases} n_i^s, & \text{for } i \in [1, I_u) \bigcup [I_d - 1, I_h - 1) \\ n_i^s - \left(\widehat{K}_i - K_i\right), & \text{for } i \in [I_u, I_d - 1) \end{cases} \tag{37}$$

where the initial value is $n_1^s = 0$. The frequency index K_i is predetermined between the transmitter and receiver, depending on the synchronization stage which can be expressed as

$$K_i = \begin{cases} a_0, & \text{for } i \in [I_u, I_F) \\ a_{i-I_F+1}, & \text{for } i \in [I_F, I_d) \\ (N - a_0) \bmod N, & \text{for } i \in [I_d, I_h) \end{cases} \tag{38}$$

where I_F, I_d, and I_h denote the start flags for FSS detection, downchirp detection and header demodulation, respectively.

A shift function is defined to describe the practical shift between $\widehat{K}_i$ and K_i for $i > I_u$, which can be expressed as

$$\gamma\left(\widehat{K}_i, K_i\right) = \begin{cases} \widehat{K}_i - K_i, & \text{for } \left|\widehat{K}_i - K_i\right| < N/2 \\ \widehat{K}_i - K_i + \varepsilon N, & \text{for } \left|\widehat{K}_i - K_i\right| \geq N/2 \end{cases} \tag{39}$$

where ε is the cycle control factor, given by

$$\varepsilon = -(-1)^{\psi(K_i/N)}. \tag{40}$$

The shift function can be used to reflect the coupling degree between SCFO and STO in the subsequent upchirp detection. The coupling is considered complete under

$$\left|\gamma\left(\widehat{K}_i, a_0\right)\right| \leq \Delta_1 \tag{41}$$

where Δ_1 is the maximum coupling error. After achieving the time-frequency coupling, the residual STO is given by

$$K'_\tau = K'_{\tau_\alpha} + K_{\tau_\beta} = -K_{f_\alpha} - \psi\left(K_{f_\beta} + K_{\tau_\beta}\right) + K_{\tau_\beta} \tag{42}$$

where the sampling interval $1/B$ is omitted.

Due to the picket-fence effect caused by the N-point DFT, the fractional part of $K_f + K'_\tau$, i.e., $K_{f_\beta} + K_{\tau_\beta} - \psi(K_{f_\beta} + K_{\tau_\beta})$, in the presence of DCFO can be estimated through [30]

$$f_{i+1}^e = f_i^e + \frac{1}{2} \cdot \frac{\left|Y_i\left(\widehat{K}_i^R\right)\right| - \left|Y_i\left(\widehat{K}_i^L\right)\right|}{2\left|Y_i\left(\widehat{K}_i\right)\right| - \left|Y_i\left(\widehat{K}_i^L\right)\right| - \left|Y_i\left(\widehat{K}_i^R\right)\right|} \tag{43}$$

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where $\widehat{K}_i^R = (\widehat{K}_i+1) \bmod N$ and $\widehat{K}_i^L = (\widehat{K}_i-1) \bmod N$ are the adjacent frequency indexes of $\widehat{K}_i$. The estimation value f_i^e is defined as the *frequency correction factor* whose initial vale is $f_0^e = 0$. This factor is iteratively calculated during the upchirp detection to correct the fraction part of $K_f + K'_\tau$ and remains unchanged during the FSS and downchirp detection. The chirp signal in the subsequent preamble detection is corrected using f_i^e which can be described as

$$r'_i(n) = r_i(n)\exp\left(-\frac{j2\pi n}{N} f_i^e\right), \text{ for } i > I_u. \tag{44}$$

2) Stage2 (FSS Detection): The purpose of the FSS detection is to verify the matching of the received network ID. After achieving the time-frequency coupling, the first FSS is considered detected under

$$\left|\gamma\left(\widehat{K}_{I_F}, a_0\right)\right| > \Delta_1, \ \left|\gamma\left(\widehat{K}_{I_F}, a_1\right)\right| \le \Delta_2 \tag{45}$$

where Δ_2 is the maximum matching error for FSS detection. The subsequent $m_1 - 1$ FSSs are then demodulated and the data frame is allowed to be received under

$$\left|\gamma\left(\widehat{K}_i, K_i\right)\right| \le \Delta_2, \text{ for } i \in [I_F, I_F + m_1 - 1] \tag{46}$$

which indicates that the transmitter's network ID is matched with the receiver. Conversely, if the network ID does not match, the data frame is discarded and the receiver restarts the upchirp detection process.

3) Stage3 (Downchirp Detection): The purpose of the downchirp detection is to estimate the integer parts of SCFO and residual STO. In the I_dth detection, the first downchirp can be expressed as

$$r_{I_d}(n) = x_{\text{down}}(n,0)e^{j2\pi\left(\frac{K_f - K'_\tau}{N} n + K_{\tau_\beta} u\left(n - N'_b\right)\right)} + w(n) \quad \text{for } n \in [0, N-1] \tag{47}$$

where $N'_b = N - \lfloor K'_\tau \rfloor$ represents the boundary between two consecutive downchirps and the additional phase $\varphi_d = -\pi(\mu\tau' - B)\tau'$ caused by residual STO is omitted. The downchirp described in (47) is derived under a positive residual STO. By demodulating this downchirp, the obtained frequency index of the spectral peak, without considering AWGN, can be expressed as

$$\begin{aligned}\widehat{K}_{I_d} &= \left(\psi\left(K_f - K'_\tau - f_{I_d}^e\right)\right) \bmod N \\ &= \left(2K_{f_\alpha} + 2\psi\left(K_{f_\beta} + K_{\tau_\beta}\right) + \psi\left(-2K_{\tau_\beta} - \Delta_e\right)\right) \bmod N\end{aligned} \tag{48}$$

where $f_{I_d}^e = K_{f_\beta} + K_{\tau_\beta} - \psi(K_{f_\beta} + K_{\tau_\beta}) + \Delta_e$ is the estimation for the fractional part of $K_f + K'_\tau$ and $\Delta_e \in (-0.5, 0.5)$ is the residual estimation error. The practical shift between $\widehat{K}_{I_d}$ and K_{I_d} is $\gamma(\widehat{K}_{I_d}, K_{I_d}) = 2K_{f_\alpha} + 2\psi(K_{f_\beta} + K_{\tau_\beta}) + \psi(-2K_{\tau_\beta} - \Delta_e)$ where the practical shift range is $[-N/2, N/2)$. The shift function is utilized to estimate K_{f_α} and K'_{τ_α} which can be described as

$$\begin{aligned}\widehat{K}_{f_\alpha} &= -\widehat{K'}_{\tau_\alpha} = \psi\left(\frac{\gamma\left(\widehat{K}_{I_d}, a_0\right)}{2}\right) \\ &= K_{f_\alpha} + \psi\left(K_{f_\beta} + K_{\tau_\beta}\right) + \psi\left(\frac{1}{2}\psi\left(-2K_{\tau_\beta} - \Delta_e\right)\right)\end{aligned} \tag{49}$$

where the maximum estimation range for SCFO is $[-B/4, B/4)$, without considering the estimation error terms. Here, the second term $\psi(K_{f_\beta} + K_{\tau_\beta})$ leads to an estimation error for K_{f_α}, whereas the last term, specifically described as

$$\psi\left(\frac{\psi\left(-2K_{\tau_\beta} - \Delta_e\right)}{2}\right) = \begin{cases} 0, & \text{for} -2K_{\tau_\beta} - \Delta_e \in (-1.5, 0.5) \\ 1, & \text{for} -2K_{\tau_\beta} - \Delta_e \in [0.5, 1.5) \end{cases} \tag{50}$$

will lead to estimation errors for both K_{f_α} and K'_{τ_α} under $-2K_{\tau_\beta} - \Delta_e \in [0.5, 1.5)$. But both the estimation errors can be counteracted in the subsequent chirp demodulation.

Apart from the last protective signal, the subsequent $m_2 - 2$ downchirp detections are utilized to estimate K_{f_α} and K'_{τ_α} to mitigate the impact of AWGN on the estimation. Accordingly, the estimation for K_{f_α} and K'_{τ_α} is refined to

$$\widehat{K}_{f_\alpha} = -\widehat{K'}_{\tau_\alpha} = \frac{1}{m_2 - 1} \sum_{i=I_d}^{I_d + m_2 - 2} \psi\left(\frac{\gamma\left(\widehat{K}_i, a_0\right)}{2}\right). \tag{51}$$

The timing factor and frequency correction factor are adjusted before the header demodulation to correct the integer parts of SCFO and STO, given by

$$n_{I_h}^s = n_{I_h - 1}^s - \widehat{K'}_{\tau_\alpha} + N/4, \ f_{I_h}^e = f_{I_h - 1}^e + \widehat{K}_{f_\alpha}. \tag{52}$$

D. LRO-Based DCFO Tracking and Demodulation

As depicted in Fig. 7, the chirp signals corresponding to the header and payload are demodulated after the synchronization. The demodulation of the chirp signal is processed by an N-point sliding demodulation window with step length N. The demodulation method outlined in (22)–(25) is applied to the chirp signal to obtain the frequency index of the spectral peak. The demodulated frequency index is then processed by LRO to recover the transmitted symbol according to the rule as follows:

$$\widehat{S}_i = \left\lfloor \frac{\widehat{K}_i}{l} \right\rceil, \text{ for } i \ge I_h. \tag{53}$$

Here, the recovery process can correct partial frequency index demodulation errors caused by residual STO, CFO (including DCFO) and AWGN, as depicted in Fig. 5.

The LRO operation increases the frequency interval between adjacent valid frequency indexes. Within this expanded frequency interval, the $l - 1$ invalid frequency indexes can be used to track DCFO by following the direction from the invalid frequency index demodulated from (25) to the nearest valid index. This tracking strategy can be specifically described as

$$f_{i+1}^e = \begin{cases} f_i^e + \rho, & \text{for } \widehat{K}_i \in \left(K_i, K_i + \left\lceil \frac{l-1}{2} \right\rceil\right] \\ f_i^e, & \text{for } \widehat{K}_i = K_i \\ f_i^e - \rho, & \text{for } \widehat{K}_i \in \left[K_i - \left\lfloor \frac{l-1}{2} \right\rfloor, K_i\right) \end{cases} \text{for } i \ge I_h \tag{54}$$

where $\rho \in (0, 1)$ denotes the fixed adjustment factor and $K_i = l \cdot \widehat{S}_i + 1$ is the nearest valid frequency index to $\widehat{K}_i$. The iterative frequency correction factor obtained from (54) is consistently applied in the subsequent chirp demodulation to track DCFO.

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The maximum correctable DCFO, denoted as f_a^*, correlates with the fixed adjustment factor, bandwidth and spreading factor, and can be expressed as

$$f_a^* \in \left[-\rho \cdot \left(\frac{B}{2^{\mathrm{SF}}}\right)^2, \rho \cdot \left(\frac{B}{2^{\mathrm{SF}}}\right)^2\right]. \tag{55}$$

However, in the presence of index demodulation errors, the practical tracking capability of the proposed strategy is inferior to that in (55).

Two types of frequency index demodulation errors can lead to the failure of DCFO tracking. During the ith demodulation, the chirp signal is assumed contaminated with positive DCFO, i.e., $f_a > 0$, but it is incorrectly demodulated as $\widehat{K}_i = K_i - 1$ caused by AWGN. This error is opposite to the practical drift direction and is therefore classified as reversed demodulation error (RDE). According to the proposed tracking strategy, RDE will lead to a reversed DCFO correction. The persistent positive DCFO and reversed correction for it can potentially cause the next chirp signal to be incorrectly demodulated as $\widehat{K}_{i+1} = K_{i+1} + 2$, which exceeds the effective correction range of LRO with $l = 3$. This type of error is defined as uncorrectable demodulation error (UDE). When $|\widehat{K} - K| < l$, UDEs include $\widehat{K} = K \pm 2$ for $l = 3$ and $\widehat{K} = K - 2$ for $l = 4$. Notably, UDE caused by the above scenarios will also lead to the reversed DCFO correction. The consecutive reversed DCFO tracking operations will result in the failure of DCFO tracking. Consequently, the practical tracking capability is correlated with the LRO factor and bit SNR, apart from the parameters described in (55). The tracking capability of the proposed strategy with varying LRO factors is assessed in Section IV.

E. CRC-Assisted Decoding Scheme

As depicted in Fig. 7, decoding is the final stage of the proposed receiver, aimed at correcting the transmission errors introduced by the channel.

The transmitted symbol, recovered from (53), is converted into a 16-bit sequence, denoted as $\widehat{\boldsymbol{\mu}'}_i = (\widehat{\mu'}_{i,0}, \widehat{\mu'}_{i,1}, \ldots, \widehat{\mu'}_{i,15})$ and then decoded according to the gray decoding rule as follows:

$$\widehat{\boldsymbol{\mu}}_i(\upsilon) = \begin{cases} \widehat{\boldsymbol{\mu}}_{i,\upsilon} = \left(\widehat{\boldsymbol{\mu}'}_{i,\upsilon-1} + \widehat{\mu'}_{i,\upsilon}\right), & \text{for } \upsilon \neq 0 \\ \widehat{\boldsymbol{\mu}}_{i,0} = \widehat{\mu'}_{i,0}, & \text{for } \upsilon = 0 \end{cases} \tag{56}$$

where $\widehat{\boldsymbol{\mu}}_i = (\widehat{\boldsymbol{\mu}}_{i,0}, \widehat{\boldsymbol{\mu}}_{i,1}, \ldots, \widehat{\boldsymbol{\mu}}_{i,15})$ is the gray decoding result. The last nonzero C_F bits of $\widehat{\boldsymbol{\mu}}_i$ are retained, denoted as $\widehat{\mathbf{e}}_i$, and written into the deinterleaving block row-wise. If $l = 3$ is adopted in the proposed system, three consecutive M-ary transmitted symbols recovered from (53) are converted into a $\lfloor \log_2(2^{3\mathrm{SF}}/27) \rfloor$-bits sequence, which is then directly written into the deinterleaving block row-wise.

The Gray-decoded bits in the deinterleaving block, denoted as $\mathbf{X}'$, are distributed as follows:

$$\mathbf{X}' = \begin{pmatrix} \widehat{\mathbf{e}}_i \\ \widehat{\mathbf{e}}_{i+1} \\ \vdots \\ \widehat{\mathbf{e}}_{i+3+C_B} \end{pmatrix} = \begin{pmatrix} \widehat{e}_{i,1} & \widehat{e}_{i,2} & \cdots & \widehat{e}_{i,C_F} \\ \widehat{e}_{i+1,1} & \widehat{e}_{i+1,2} & \cdots & \widehat{e}_{i+1,C_F} \\ \vdots & \vdots & \ddots & \vdots \\ \widehat{e}_{i+3+C_B,1} & \widehat{e}_{i+3+C_B,2} & \cdots & \widehat{e}_{i+3+C_B,C_F} \end{pmatrix}. \tag{57}$$

These bits are then deinterleaved according to the rule

$$\mathbf{X}(\xi, \eta) = \mathbf{X}'(3 + C_B - \eta, C_F - 1 - ((\xi + \eta + 2) \bmod C_F)) \quad \text{for } \xi \in [0, C_F - 1],\ \eta \in [0, 3 + C_B] \tag{58}$$

where $\mathbf{X}$ denotes the deinterleaved block and ξ, η are the row and column indexes, respectively. Each row of the deinterleaved block is read to form a codeword, denoted as $\widehat{\mathbf{d}}_k$ where k is the codeword index. The codeword is then added with the whitening sequence $\mathbf{w}_k$ in GF(2). The dewhitening process is described as

$$\widehat{\mathbf{c}}_k = \left(\widehat{\mathbf{d}}_k + \mathbf{w}_k\right) \bmod 2 = \left(\widehat{c}_{k,0}, \widehat{c}_{k,1}, \ldots, \widehat{c}_{k,3+C_B}\right) \tag{59}$$

where $\mathbf{w}_k$ is the whitening sequence utilized in (5), and $\widehat{\mathbf{c}}_k$ is the kth dewhitened codeword.

In the channel decoding, we propose leveraging (5, 4) Hamming codes to identify the erroneous codewords, and traversing the finite potential error patterns to correct the transmission errors. The Hamming decoding is performed on the dewhitened codeword obtained from (59) by multiplying it with the parity matrix. The decoding process is described as follows:

$$\begin{aligned} \widehat{\mathbf{B}'}_k &= \left(\widehat{c}_{k,0}, \widehat{c}_{k,1}, \widehat{c}_{k,2}, \widehat{c}_{k,3}, \widehat{c}_{k,4}\right) \cdot \mathbf{H}_1 \\ &= \left(\widehat{c}_{k,0}, \widehat{c}_{k,1}, \widehat{c}_{k,2}, \widehat{c}_{k,3} \middle| \sum_{i=0}^{4} c_{k,i}\right) \\ &= \left(\widehat{\mathbf{b}'}_k \middle| \widehat{r}_k\right) \end{aligned} \tag{60}$$

where $\mathbf{H}_1$ denotes the parity matrix with $C_B = 1$ and $\widehat{r}_k$ is the parity bit. Through the Hamming decoding, we can obtain the codeword sequence denoted as $\widehat{\mathbf{b}'}$ and the parity-bit sequence denoted as $\widehat{\mathbf{r}}$. The parity-bit sequence is utilized to instruct the indexes of erroneous codewords, which can be described as

$$\mathbf{k} = \{k | \widehat{r}_k = 1, k = 0, 1, 2, \ldots\} \tag{61}$$

where $\mathbf{k}$ is the index set of erroneous codewords with size of Z. These indexes are then used to calculate the potential error pattern indexes as described in (70).

The codeword sequence generated from (60) can be divided into the information bits and the data domain CRC, which can be expressed as

$$\widehat{\mathbf{b}'} = \left(\widehat{b'}_0, \widehat{b'}_1, \ldots, \widehat{b'}_{L-1} | \widehat{b'}_L, \widehat{b'}_{L+1}, \ldots, \widehat{b'}_{L+15}\right) = \left(\widehat{\mathbf{b}} \mid \widehat{\mathbf{p}}\right). \tag{62}$$

The polynomials of these two parts are given as follows:

$$\widehat{B}(X) = \sum_{i=0}^{L-1} \widehat{b'}_i X^i, \widehat{P}(X) = \sum_{i=0}^{15} \widehat{b'}_{L+i} X^i. \tag{63}$$

The checking process can be described as

$$\Delta P(X) = \widehat{B}(X) \bmod g_{\mathrm{crc16}}(X) - \widehat{P}(X) = \sum_{i=0}^{15} \Delta p_i X^i \tag{64}$$

where Δp_i denotes the CRC parity bit. The nonzero CRC parity bits indicate transmission errors.

The error patterns used to correct the transmission errors comprise the data domain CRC part and the information part.

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There are sixteen error patterns for the data domain CRC part, which can be described as

$$\mathbf{f} = \begin{pmatrix} \mathbf{f}_0 \\ \mathbf{f}_1 \\ \vdots \\ \mathbf{f}_{15} \end{pmatrix} = \begin{pmatrix} 0 & \cdots & 0 & 1 \\ 0 & \cdots & 1 & 0 \\ \vdots & \iddots & \vdots & \vdots \\ 1 & \cdots & 0 & 0 \end{pmatrix}_{16\times 16} \tag{65}$$

where $\mathbf{f}_\nu$ denotes the νth error pattern with $f_{\nu,15-\nu} = 1$. The polynomial of the error pattern can be expressed as

$$F_\nu(X) = \sum_{i=0}^{15} f_{\nu,i} X^i = X^{15-\nu}, \text{ for } \nu \in [0, 15] \tag{66}$$

where $f_{\nu,i}$ denotes the ith element of $\mathbf{f}_\nu$. The single-bit error vectors for the information part are given as follows:

$$\mathbf{U} = \begin{pmatrix} \mathbf{u}_0 \\ \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_{L-1} \end{pmatrix} = \begin{pmatrix} 0 & \cdots & 0 & 1 \\ 0 & \cdots & 1 & 0 \\ \vdots & \iddots & \vdots & \vdots \\ 1 & \cdots & 0 & 0 \end{pmatrix}_{L\times L} \tag{67}$$

where $\mathbf{u}_n$ denotes the nth error vector with $u_{n,L-1-n} = 1$. The CRC polynomial of $\mathbf{u}_n$ is generated as follows:

$$R_n(X) = U_n(X) \bmod g_{\text{crc16}}(X) = \sum_{i=0}^{15} r_{n,i} X^i \tag{68}$$

where $U_n(X) = X^{L+15-n}$ is the polynomial of $\mathbf{u}_n$. The overall error patterns accordingly can be described as

$$F_\nu(X) = \begin{cases} X^{15-\nu}, & \text{for } \nu \in [0, 15] \\ R_{\nu-16}(X), & \text{for } \nu \in [16, L+15]. \end{cases} \tag{69}$$

The indexes of four potential error patterns for the specified erroneous codeword are provided as follows:

$$\nu_{z,\zeta_z} = L + 15 - 4 \cdot k_z - \zeta_z, \text{ for } z \in [0, Z), \ \zeta_z \in [0, 3] \tag{70}$$

where k_z is the zth element of the index set described in (61) and ζ_z denotes the bit index in the k_zth erroneous codeword. The CRC-assisted decoding is performed by traversing the potential error patterns as described in (69) and (70) and uses the combination summation of them in GF(2) to locate the error bits. When the traversal summation result of these error patterns satisfies

$$\sum_{z=1}^{Z} F_{\nu_{z,\zeta_z}}(X) = \Delta P(X), \text{ for } \zeta_0, \zeta_1, \ldots, \zeta_Z \in [0, 3] \tag{71}$$

the indexes of the error bits are therefore determined and the transmission errors can be effectively corrected. By comparing 4^Z error patterns, up to Z error bits can be corrected. Note that, not all Z error bits can be corrected since the error patterns may be the same with different error combinations, especially for large Z.

TABLE II
ESSENTIAL PARAMETERS

Variable	Parameters
Spreading factor	$SF = 10, 12$
Bandwidth	$B = 125$ kHz
Payload length	$P_L = 128$ bytes
Coding rate	$R = 4/5$
Upchirps number	$m_0 = 7$
FSSs number	$m_1 = 2$
Downchirps number	$m_2 = 3$
LRO factor	$l = 1, 3, 4$
Fixed adjustment factor	$\rho = 0.2$

F. Complexity Analysis of the Receiver

The proposed synchronization algorithm, LRO-based DCFO tracking and demodulation are based on the chirp demodulation outlined in (22)–(25), with limited arithmetic operations. The complexity of chirp demodulation is $O(N \log_2 N)$, dominated by the N-point FFT. The synchronization algorithm introduces $2N$ complex multiplications for fractional offset compensation and threshold detection, and minor arithmetic operations to estimate SCFO and STO, with complexity of $O(N_1 \cdot N)$. The LRO-based DCFO tracking strategy necessitates one remainder operation and one addition to update the frequency correction factor as described in (54), and N complex multiplications to correct the DCFO, whose complexity is $O((N_2+N_3)\cdot N)$. The computation complexity of the proposed receiver is accordingly $O(N_s N(\log_2 N+1)) \rightarrow O(N_s N \log_2 N)$.

IV. SIMULATION RESULTS

In this section, we evaluate the performance of the proposed chirp modulation system. The LRO-based chirp modulation with expanded frequency interval and the chirp modulation without LRO are assessed in the presence of SCFO and DCFO, respectively. Simulation results show that the LRO-based chirp modulation is feasible for LEO satellite communication. The system's performance under different CFO conditions are then evaluated, revealing that the proposed system is insensitive to both SCFO and DCFO. The CRC-assisted decoding scheme and gray coding are also evaluated across different ranges of DCFO. Finally, we evaluate the DCFO tracking capability of the LRO-based tracking strategy with different spreading factors and LRO factors. Essential parameters of the proposed system adopted in the simulations are outlined in Table II.

A. Performance Under Different Ranges of SCFO and DCFO

The performance of the LRO-based chirp modulation with $l = 4$ and the chirp modulation without LRO are compared under two SCFO ranges, i.e., $[-B/10, B/10]$ and $[-B/5, B/5]$. For simplicity, the chirp modulation without LRO is represented as $l = 1$. The simulation results are illustrated in Fig. 8.

The synchronization algorithm proposed in this work is designed for LEO satellite communication scenarios characterized by high-dynamic Doppler offset. The estimator for the fractional CFO in [24] and [25] is ineffective in the presence

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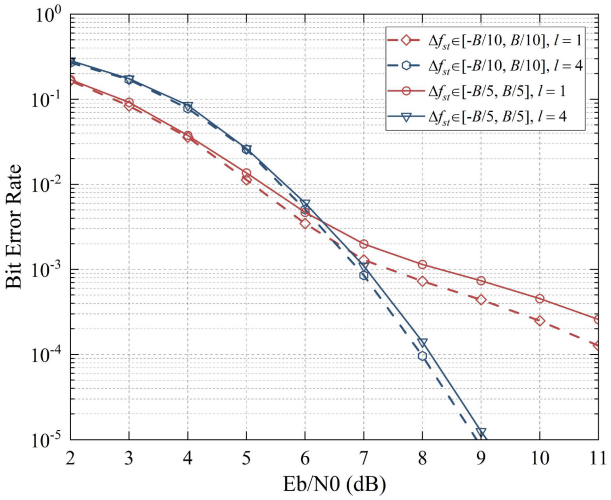


Fig. 8. BERs of the proposed system with the LRO-based chirp modulation ($l = 4$) and the chirp modulation without LRO ($l = 1$) under two ranges of SCFO where Δf_{st} denotes SCFO.

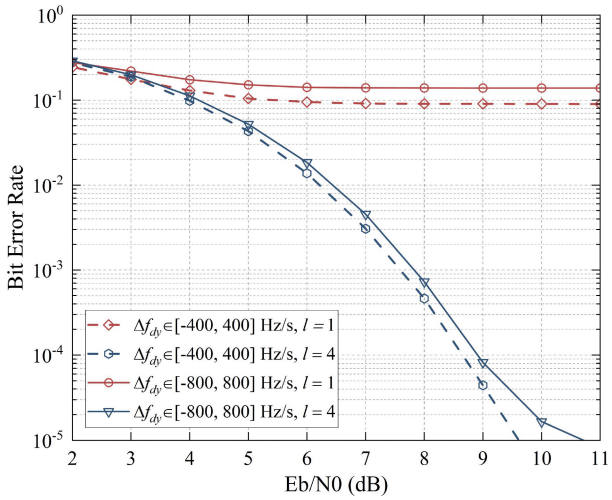


Fig. 9. BERs of the proposed system with the LRO-based chirp modulation ($l = 4$) and the chirp modulation without LRO ($l = 1$) under two ranges of DCFO where Δf_{dy} represents the DCFO.

of DCFO. This article employs the estimator described in (43) to handle the combined value of fractional SCFO and STO, which is then corrected through frequency compensation. Consequently, the fractional STO is not realigned in our work. The proposed LRO can effectively mitigate the impact of the residual fractional STO on chirp demodulation. However, the performance of the chirp modulation without LRO is degraded by the residual fractional STO.

As illustrated in Fig. 8, the chirp modulation without LRO exhibits linear BERs for $E_b/N_0 \geq 7$ dB under the two SCFO ranges. In contrast, the BERs of the LRO-based chirp modulation decrease rapidly as E_b/N_0 increases. When the SCFO range is doubled to $[-B/5, B/5]$, the E_b/N_0 required to achieve $P_e = 10^{-5}$ increases by only 0.1dB, demonstrating that the LRO-based chirp modulation is insensitive to SCFO. The simulation results also illustrate that the synchronization algorithm proposed in this work can effectively address SCFO and integer STO, while the residual fractional STO has minimal impact on the LRO-based chirp modulation.

The evaluation of the two modulation schemes under two ranges of DCFO, i.e., $[-400, 400]$ and $[-800, 800]$ Hz/s, is illustrated in Fig. 9. These DCFO ranges cover the maximum Doppler drift rate for LEO satellite communication an altitude of 560km with carrier frequencies of 868 MHz and 2.1 GHz, respectively [4]. From Fig. 9, we can find that the chirp modulation without LRO fails to operate effectively in the presence of DCFO due to its lack of tracking capability, as evidenced by the flat BERs for both DCFO ranges. In contrast, the BER of the improved chirp modulation with LRO still exhibits a rapid decline under $[-400, 400]$ Hz/s, confirming the feasibility of the proposed system in the presence of DCFO.

However, a slight error floor is observed in the BER of the improved chirp modulation with $l = 4$ under $[-800, 800]$ Hz/s when $E_b/N_0 \geq 9$ dB. The E_b/N_0 required to achieve $P_e = 10^{-5}$ increases by 1.1 dB compared to that under $[-400, 400]$ Hz/s. The observed phenomenon is attributed to

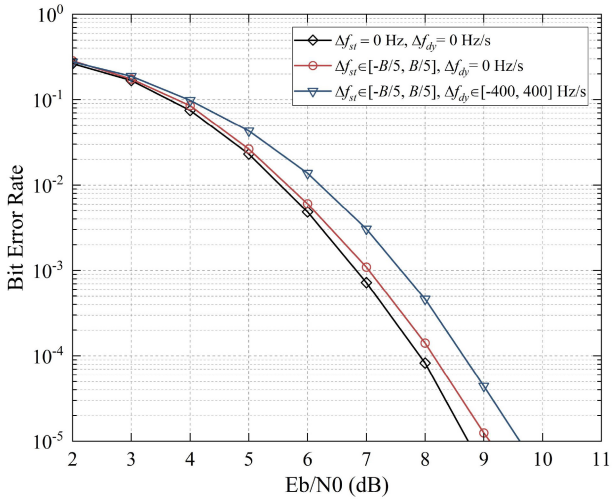


Fig. 10. BERs of the proposed system under three CFO conditions: Without CFO, with SCFO, and with DCFO.

the increase in UDE caused by the reversed DCFO tracking originating from RDEs as analyzed in Section III, especially under higher DCFO. Nevertheless, the transmission errors caused by UDE can be effectively corrected by the proposed CRC-assisted decoding scheme, as evidenced by the rapid decline in BER under the same conditions illustrated in Fig. 11.

The proposed system is then evaluated under three different CFO conditions: 1) without CFO; 2) with SCFO; and 3) with DCFO. The evaluation results are illustrated in Fig. 10. The BERs under the three CFO conditions exhibit similarly rapid decline as E_b/N_0 increases. The black line denotes the ideal BER in the absence of CFO, where the E_b/N_0 required to achieve $P_e = 10^{-5}$ is 8.75 dB. In the presence of SCFO, the required E_b/N_0 increases by 0.3 dB, and in the presence of DCFO, it increases by 0.8 dB, compared to the ideal BER performance. The former is attributed to the energy leakage of the chirp signal caused by the fractional SCFO, while

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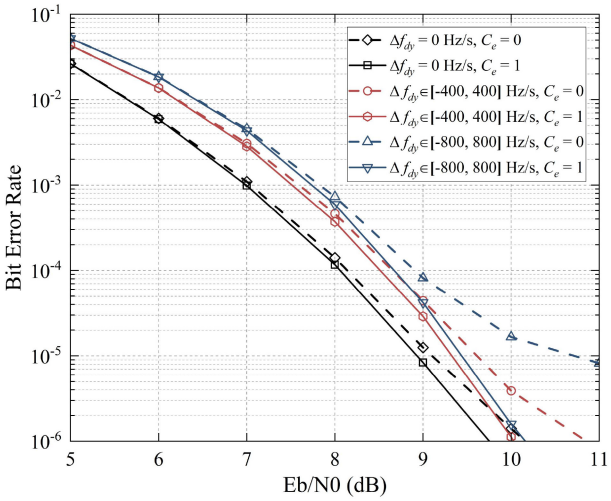


Fig. 11. BERs of the CRC-assisted decoding scheme under three DCFO ranges where C_e denotes the CRC-assisted decoding option.

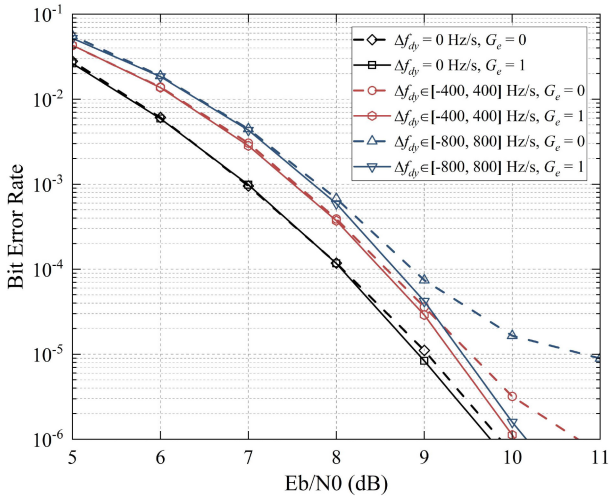


Fig. 12. BERs of gray coding under three DCFO ranges where G_e denotes the gray coding option.

the latter is caused by UDEs originating from the reversed DCFO tracking as previously analyzed. The simulation results also confirm that the proposed system, employing the LRO-based chirp modulation with expanded frequency interval, is insensitive to both SCFO and DCFO.

B. Performance of CRC-Assisted Decoding and Gray Coding

The performance of the CRC-assisted decoding scheme is evaluated under three DCFO ranges: 1) 0; 2) [−400, 400]; and 3) [−800, 800] Hz/s. The evaluation results are illustrated in Fig. 11. It is observed that the coding gain in E_b/N_0 required to achieve $P_e = 10^{-5}$ increases with the DCFO range. The coding gain in the absence of DCFO is approximately 0.17 dB, while an improvement of 0.3 dB is observed under [−400, 400] Hz/s and 1.3 dB under [−800, 800] Hz/s. Furthermore, the slight error floor observed in Fig. 9 is eliminated by the CRC-assisted decoding, attributed to the generalized coding scheme. Specifically, gray coding reduces the bit errors caused by UDEs, and the diagonal interleaving disperses the consecutive error bits, which maximize the correction capability of the CRC-assisted decoding. When the required BER is reduced to 10^{-6}, the coding gain is further amplified.

We then evaluate the performance of gray coding under three DCFO ranges, as illustrated in Fig. 12. It is observed that the gray coding gain in E_b/N_0 increases with both the DCFO range and the required BER. When the required BER is 10^{-5}, the gray coding gain is approximately 0.1 dB in the absence of DCFO, whereas 0.2 dB under [−400, 400] Hz/s and 1.4 dB under [−800, 800] Hz/s. Notably, the CRC-assisted decoding is employed in the evaluation of gray coding. The gray coding gain observed in Fig. 12 is approximately equal to that of the CRC-assisted decoding, indicating that the error correction capability of the proposed CRC-assisted decoding is based on the reduction of error bits, which is attributed to gray coding. Accordingly, gray coding is indispensable in the proposed generalized coding scheme which employs the (5, 4) Hamming codes and CRC-assisted decoding.

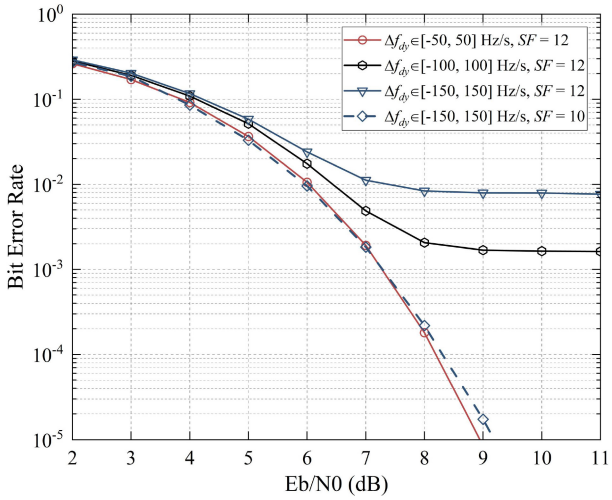


Fig. 13. BERs of the proposed system with SF = 10 and 12 under different DCFO ranges where $C_e, G_e = 1$ and $l = 4$.

C. Performance Under Different Spreading Factors

To confirm the practical DCFO tracking capability of the proposed strategy under specified spreading factor, we assess the performance of the proposed system with SF = 10 and 12 under different DCFO ranges. The results are illustrated in Fig. 13. According to (55), the maximum correctable DCFO under SF = 12 is [−186, 186] Hz/s. However, an error floor is observed in the BER of the proposed system with SF = 12 for both the DCFO range of [−150, 150] and [−100, 100] Hz/s. When the DCFO range is narrowed to [−50, 50] Hz/s, the BER exhibits a rapid decline, similar to that under SF = 10 for the DCFO range of [−150, 150] Hz/s.

Both the two error floors result from that the adopted DCFO ranges in simulation exceed the practical tracking capability of the proposed DCFO tracking strategy, which is inferior to the maximum correctable DCFO written in (55), attributed to the index demodulation errors caused by AWGN and timing error. Furthermore, we find that the tracking capability is severely degraded as the spreading factor increases. The proposed

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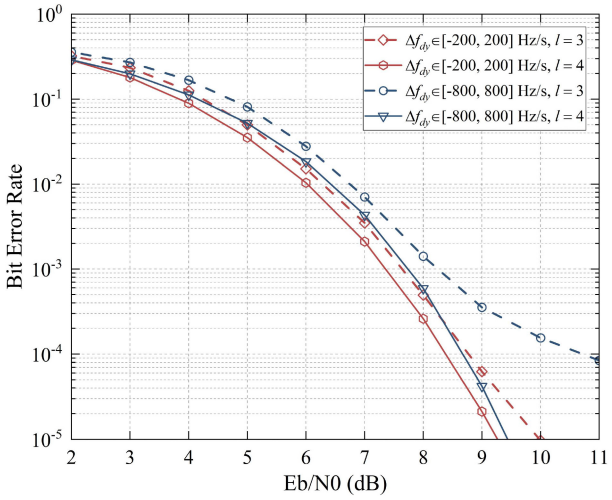


Fig. 14. BERs of the proposed system with $l = 3$ and $l = 4$ under two DCFO ranges where $C_e, G_e = 1$.

tracking strategy with SF = 10 can effectively track DCFO within [−800, 800] Hz/s, which covers the maximum DCFO at sub-GHz carrier frequencies in LEO satellite communication. Both the tracking capabilities under SF = 10 and SF = 12 are significantly smaller than the maximum correctable DCFO. Therefore, for a given bandwidth and fixed adjustment factor, the spreading factor should be selected with caution to achieve successful frequency tracking in LEO satellite communication.

D. Performance Under Different LRO Factors

In the proposed LRO-based chirp modulation, we propose to select the optimal LRO factor as $l = 3$ to improve the spectral efficiency. The chirp modulation with $l = 3$ utilizes the M-ary numbering system for symbol mapping where gray coding cannot be effectively applied. Besides, the symbol mapping method is susceptible to symbol demodulation errors. The correct recovery of the bit sequence at the receiver relies on the accurate demodulation of three consecutive symbols, particularly the first two symbols. Considering the absence of gray coding, the symbol mapping method can lead to multiple consecutive bit errors despite in the presence of single symbol demodulation error.

Furthermore, the proposed tracking strategy with $l = 3$ is more susceptible to reversed DCFO tracking compared to that with $l = 4$ due to the narrowed correction range for the index demodulation error. This susceptibility amplifies the drawbacks of the symbol mapping method. To validate the above analysis, we evaluate the performance of the proposed system with $l = 3$ and $l = 4$ under two DCFO ranges, as illustrated in Fig. 14.

A slight error floor is observed in the BER for $l = 3$ under the DCFO range of [−800, 800] Hz/s, while the BER for $l = 4$ maintains a rapid decline under the same range. The difference in E_b/N_0 to achieve $P_e = 10^{-4}$ between the two LRO factors is approximately 2 dB for the DCFO range of [−800, 800] Hz/s. These simulation results validate the above analysis. When the DCFO range is narrowed to [−200, 200] Hz/s, the BERs under the two LRO factors exhibit similarly rapid declines, with a difference of 0.4 dB in E_b/N_0 for $P_e = 10^{-4}$ and 0.74 dB for $P_e = 10^{-5}$. The simulation results demonstrate that the adverse impact of $l = 3$ on the system performance is mitigated as the DCFO range narrows. The performance degradation under the DCFO range of [−200, 200] Hz/s is acceptable in exchange for the improved spectral efficiency, achieved by transmitting an additional bit for every three consecutive symbols, which is accordingly a tradeoff between the system BER performance and spectral efficiency. We can select larger LRO factors for reliability requirements, i.e., $i \geq 4$, while smaller LRO factor is preferred to enhance the data rate.

V. Conclusion

In this work, we propose an effective chirp modulation communication system design for LEO satellite IoT. The proposed LRO-based chirp modulation scheme with specified frequency interval effectively enhances the robustness of the proposed system against residual CFO and STO. The proposed optimal LRO factor improves the spectral efficiency, at the cost of an acceptable system performance degradation. The low-complexity synchronization algorithm and LRO-based tracking strategy proposed in this article effectively address the SCFO and STO estimation, as well as the DCFO tracking. Furthermore, the CRC-assisted decoding effectively corrects the transmission errors caused by the channel. The simulation results confirm the feasibility of the proposed communication system design for LEO satellite IoT.

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